Practical:02

2.1: *Create an Android app that defines custom colors in colors.xml and applies them to the background of a TextView*

STEP 1 — Create New Project

1. Open **Android Studio**
2. Click **New Project**
3. Select **Empty Views Activity**
4. Click **Next**
5. Fill details:

   Project Name: Practical2_1_CustomColors

   Package Name: *(auto)*

   Language: Java

   Minimum SDK: API 21
6. Click **Finish**

Android separates reusable items into folders under **res/**
Examples:

- `colors.xml` → stores color definitions
- `strings.xml` → stores text
- `styles.xml` → themes
- `layout.xml` → UI structure

This practical focuses on **colors.xml**

STEP 3 — Define Custom Colors in colors.xml

Go to:

app → res → values → colors.xml

Replace everything inside with:

```
<?xml version="1.0" encoding="utf-8"?>

<resources>

   <color name="myBackground">#FF9800</color>   <!-- Orange -->

   <color name="myTextColor">#FFFFFF</color>    <!-- White -->

</resources>
```

STEP 4 — Design the UI in activity_main.xml

Go to
app → res → layout → activity_main.xml

Replace with:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center">
    <TextView
        android:id="@+id/txtSample"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Hello Custom Colors!"
        android:textSize="24sp"
        android:padding="20dp" />
</LinearLayout>
```

Why not set colors here?

Because **practical requires setting color using Java code**.

STEP 5 — Write Java code in MainActivity.java

Go to:
app → java → yourpackagename → MainActivity.java

Replace code with:

```
package com.example.practical2_1_customcolors;
import android.os.Bundle;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;
public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
```

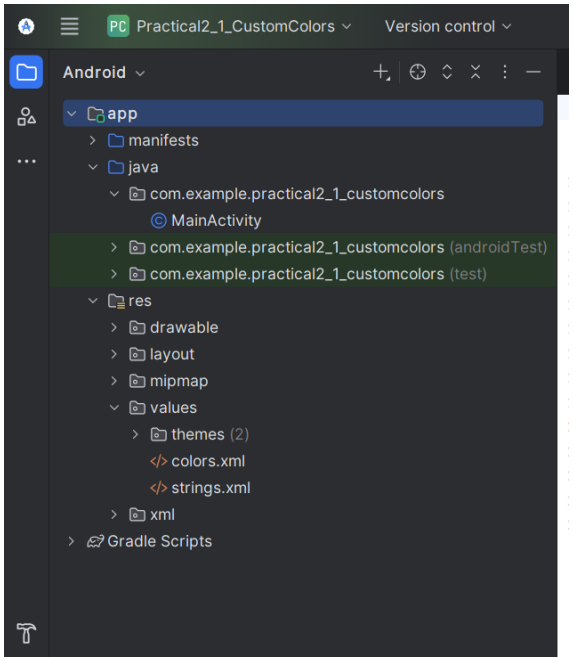
```
        setContentView(R.layout.activity_main);

        // Step 1: Find TextView by ID

        TextView txt = findViewById(R.id.txtSample);

        // Step 2: Apply custom colors from colors.xml

        txt.setBackgroundColor(getResources().getColor(R.color.myBackground));

        txt.setTextColor(getResources().getColor(R.color.myTextColor));

    }

}
```

STEP 6 — Run the App

1. Choose **Emulator** or **Physical Device**
2. Press **Run ▶**
3. Output Expected:

A screen displaying:

Orange background TextView
White text inside
Text: **Hello Custom Colors!**



Problem Statement:02

To implement an Android application using resource files to dynamically modify displayed text and background color through user interaction.

STEP 1: Create the Android Project

Open Android Studio

Click → **New Project**

Choose Template

Select:

- **Empty Views Activity**
  Click **Next**

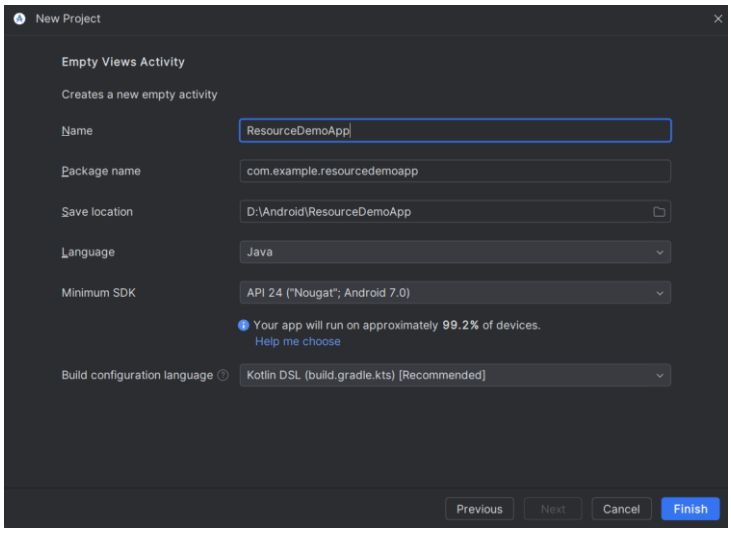
Project Configuration (IMPORTANT)

Fill exactly like this (you may change app name slightly):

- **Name:** ResourceDemoApp
- **Package name:** auto-generated (leave it)
- **Save location:** default
- **Language: Java**
- **Minimum SDK:** API 21 or above

Click **Finish**

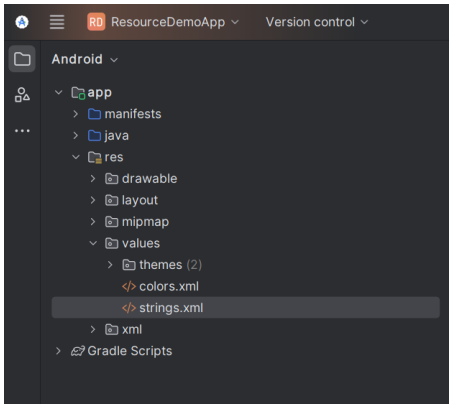
Wait till **Gradle build completes**.



Step3: **Open:**

res → values → strings.xml

(SETUP STRING RESOURCES)



Code:

```
<resources>
   <string name="app_name">ResourceDemoApp</string>
   <!-- Text shown on screen -->
   <string name="light_mode_text">Welcome to Resource Demo</string>
   <string name="dark_mode_text">Dark Mode Activated</string>
   <!-- Button labels -->
   <string name="btn_dark">Switch to Dark Mode</string>
   <string name="btn_light">Switch to Light Mode</string>
</resources>
```

Step:04

res → values → colors.xml

Code:

```
<resources>

   <color name="light_background">#FFFFFF</color>

   <color name="dark_background">#222222</color>

   <color name="light_text">#000000</color>

   <color name="dark_text">#FFFFFF</color>

</resources>
```

Step:05

res → layout → activity_main.xml

Replace everything with this:

```
<?xml version="1.0" encoding="utf-8"?>

<RelativeLayout

   xmlns:android="http://schemas.android.com/apk/res/android"

   android:id="@+id/main_layout"

   android:layout_width="match_parent"
```

```
    android:layout_height="match_parent"
    android:background="@color/light_background">
    <TextView
        android:id="@+id/tvMessage"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="@string/light_mode_text"
        android:textSize="22sp"
        android:textColor="@color/light_text"
        android:layout_centerInParent="true"/>
    <Button
        android:id="@+id/btnToggle"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="@string/btn_dark"
        android:layout_centerHorizontal="true"
        android:layout_alignParentBottom="true"
        android:layout_marginBottom="50dp"/>
</RelativeLayout>
```

Step6:Locate: java → your.package.name → MainActivity.java

Code:

```
package com.example.resourcedemoapp;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.widget.Button;
import android.widget.RelativeLayout;
import android.widget.TextView;
public class MainActivity extends AppCompatActivity {
    boolean isLightMode = true;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        RelativeLayout mainLayout = findViewById(R.id.main_layout);
        TextView tvMessage = findViewById(R.id.tvMessage);
        Button btnToggle = findViewById(R.id.btnToggle);
```

```
        btnToggle.setOnClickListener(v -> {
            if (isLightMode) {
                // Switch to Dark Mode
                mainLayout.setBackgroundColor(getResources().getColor(R.color.dark_background));
                tvMessage.setText(getString(R.string.dark_mode_text));
                tvMessage.setTextColor(getResources().getColor(R.color.dark_text));
                btnToggle.setText(getString(R.string.btn_light));
                isLightMode = false;
            } else {
                // Switch to Light Mode
                mainLayout.setBackgroundColor(getResources().getColor(R.color.light_background));
                tvMessage.setText(getString(R.string.light_mode_text));
                tvMessage.setTextColor(getResources().getColor(R.color.light_text));
                btnToggle.setText(getString(R.string.btn_dark));
                isLightMode = true;
            }

        });
    }
}
```

Now **Run the App**